Comparación del tratamiennto del cáncer de próstata localizado con radioterapia y cirugía robótica

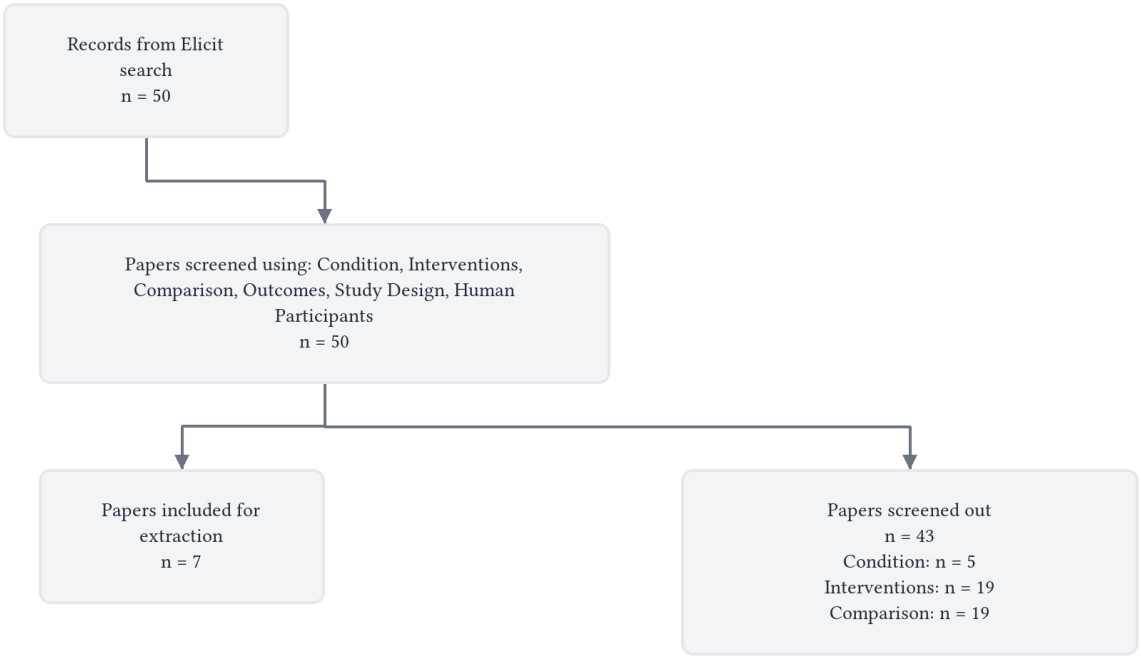
La evidencia disponible no muestra diferencias estadísticamente significativas en supervivencia libre de metástasis, control bioquímico ni mortalidad específica por cáncer entre la prostatectomía radical y la radioterapia en el cáncer de próstata localizado, aunque los resultados funcionales a corto y medio plazo son mejores con radioterapia; los datos específicos sobre cirugía robótica son insuficientes para establecer conclusiones diferenciadas respecto a la cirugía abierta.

Abstract

Across seven sources — including one meta-analysis of eight retrospective cohorts totaling 10,062 patients [1], one randomized trial of 1,643 participants [2], and several smaller retrospective studies — no statistically significant difference in oncological outcomes between radical prostatectomy and radiotherapy was consistently demonstrated for localized prostate cancer. The pooled meta-analytic estimate suggested a 21% lower risk of distant metastasis with surgery compared to external beam radiotherapy, but this was not statistically significant (HR 0.79, 95% CI 0.50–1.24; I^2 = 90.34%) [1]. Biochemical recurrence-free survival showed no significant treatment difference in low- and intermediate-risk disease at up to 7 years, though survival curves crossed between 5 and 7 years — with radiotherapy performing better early and surgery appearing more durable at 7 years [3]. Cancer-specific mortality was likewise equivalent between surgery and radiotherapy at a median 10-year follow-up in the randomized evidence (p = 0.48) [2].

The evidence base, however, cannot directly answer the specific question of robotic-assisted prostatectomy versus radiotherapy: no source isolated RARP outcomes from open surgery [1–3]. Interpretation of the comparative oncological data is further complicated by the systematic co-administration of androgen deprivation therapy with radiotherapy across all included cohorts [1], substantial heterogeneity in patient risk profiles across studies [1, 2], and the predominance of retrospective designs susceptible to selection bias [1]. In terms of functional outcomes, radical prostatectomy was associated with greater urinary incontinence, erectile dysfunction, and urinary urgency than radiotherapy at 2 and 5 years, with differences attenuating by 15 years [2]. The available evidence is therefore sufficient to support broad equivalence in oncological efficacy between the two modalities within specific risk strata, but does not permit conclusions specific to robotic surgery as a distinct intervention.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Comparación del tratamiennto del cáncer de próstata localizado con radioterapia y cirugía robótica"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Condition**: Does the study focus on patients with localized prostate cancer?
- **Interventions**: Does the study include both radiotherapy and robotic surgery (robot-assisted radical prostatectomy) as treatment interventions?
- **Comparison**: Does the study directly compare radiotherapy with robotic surgery (robot-assisted radical prostatectomy) for localized prostate cancer?
- **Outcomes**: Does the study report at least one clinically relevant outcome such as oncological outcomes (e.g., biochemical recurrence, survival), functional outcomes (e.g., urinary, sexual, or bowel function), or quality of life?
- **Study Design**: Is the study a randomized controlled trial, observational study (cohort or case-control), systematic review, or meta-analysis?
- **Human Participants**: Does the study involve human patients (not animal models, cell lines, or simulations)?

Papers that failed any criterion were automatically excluded; all other papers, including those with a Maybe on any criterion, were screened in.

At abstract screening, the number of papers excluded for each primary reason was:

- **Condition**: n = 5
- **Interventions**: n = 19
- **Comparison**: n = 19

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design**:

  Extract the methodological design of the study comparing radiotherapy vs. radical prostatectomy (including robotic) for localized prostate cancer, including:

- Study type (randomized controlled trial, retrospective cohort, prospective cohort, systematic review, meta-analysis, etc.)
- Total sample size and number of patients per treatment arm (surgery vs. radiotherapy)
- Study period (years of patient recruitment/follow-up)
- Median or mean follow-up duration in months/years
- Country or setting where the study was conducted

- **Patient Risk Profile**:

  Extract the baseline clinical characteristics and risk stratification of patients included in the comparison of surgery vs. radiotherapy for localized prostate cancer, including:

- Risk group classification (low, intermediate, high, locally advanced — per D'Amico, EAU, NCCN, or other criteria stated)
- Baseline PSA levels (mean, median, or range)
- Gleason score or ISUP grade group distribution
- Clinical TNM staging (T1, T2, T3, etc.)
- Age (mean, median, or range)
- Any exclusion criteria relevant to patient selection bias between groups

- **Treatment Details**:

  Extract the specific treatment modalities used in each arm of the comparison for localized prostate cancer, including:

For surgery:

- Type of prostatectomy (open radical, laparoscopic, robotic-assisted/RARP)
- Whether pelvic lymph node dissection was performed and extent (standard vs. extended)
- Use of nerve-sparing technique

For radiotherapy:

- Type of RT (external beam radiotherapy/EBRT, intensity-modulated/IMRT, stereotactic body/SBRT, brachytherapy, or combination)
- Total dose and fractionation scheme (if reported)
- Use of androgen deprivation therapy (ADT): yes/no, duration, and type

Note any multimodal or adjuvant treatments (e.g., salvage RT after surgery, adjuvant hormonal therapy)

- **Biochemical Recurrence**:

  Extract all data related to biochemical recurrence (BCR) comparing radical prostatectomy vs. radiotherapy for localized prostate cancer, including:

- Definition of BCR used for each arm (e.g., PSA ≥0.2 ng/mL for surgery; Phoenix criterion PSA nadir +2 ng/mL for RT)

- BCR-free survival rates at reported time points (e.g., 3, 5, 7, 10 years) for each treatment arm
- Hazard ratios (HR) with 95% confidence intervals comparing both arms
- Statistical significance (p-value)
- Any subgroup analyses by risk group (low, intermediate, high)
- Kaplan-Meier survival curves data if reported numerically

- **Survival Outcomes**:

  Extract all long-term oncological survival endpoints comparing radical prostatectomy vs. radiotherapy for localized prostate cancer, including:

- Metastasis-free survival (MFS): rates at key time points and HR with 95% CI
- Cancer-specific survival (CSS) or prostate cancer-specific mortality: rates and HR
- Overall survival (OS): rates at key time points and HR with 95% CI
- Disease-free or progression-free survival (if reported separately from BCR)
- Statistical significance for each outcome
- Whether outcomes are adjusted for confounders (and which ones)

- **Functional Outcomes**:

  Extract patient-reported or clinician-reported functional outcomes and quality of life (QoL) comparing robotic/radical prostatectomy vs. radiotherapy for localized prostate cancer, including:

- Urinary function: incontinence rates (type, severity, timeline), urinary obstruction or irritative symptoms
- Sexual/erectile function: erectile dysfunction rates, potency preservation rates, timeline of recovery
- Bowel/rectal function: rectal toxicity, proctitis (applicable mainly to RT arm)
- Validated QoL instruments used (e.g., EPIC, IPSS, IIEF, SF-36) and scores if reported
- Time points at which functional outcomes were assessed (e.g., 3, 6, 12, 24 months post-treatment)

- **Complications & Toxicity**:

  Extract adverse events, complications, and treatment-related toxicity for each arm (surgery vs. radiotherapy) in the treatment of localized prostate cancer, including:

For surgery (radical/robotic prostatectomy):

- Intraoperative and perioperative complications (e.g., bleeding, conversion rates, anastomotic leakage)
- Postoperative complications (e.g., strictures, lymphocele, Clavien-Dindo grading if used)

For radiotherapy:

- Acute and late genitourinary toxicity (grade and rates, e.g., per CTCAE or RTOG scale)
- Acute and late gastrointestinal/rectal toxicity (grade and rates)
- Toxicity related to ADT if used (e.g., metabolic, cardiovascular effects)
- Overall complication/toxicity rates per arm and severity grading system used

- **Main Comparative Findings**:

  Extract the authors' primary conclusions specifically regarding the comparison between radical prostatectomy (robotic or open) and radiotherapy for localized prostate cancer, including:

- Which treatment showed superior, equivalent, or inferior outcomes, and for which specific endpoints (biochemical control, survival, function, toxicity)
- Whether conclusions differ by risk group (low, intermediate, high)
- Limitations acknowledged by the authors that affect the interpretation of the comparison (e.g., selection bias, lack of randomization, short follow-up, heterogeneity in RT techniques or ADT use)
- Authors' recommendations for clinical practice or future research regarding treatment choice

Results

Characteristics of Included Studies

Study	Full text retrieved?	Study type	Total sample	Treatment arms	Risk group	Primary endpoint	Median follow-up	Setting
Galdós-Bejar et al., 2024 [1]	Yes	Systematic review and meta-analysis of retrospective cohorts [1]	10,062 (8 cohorts) [1]	6,414 RP vs. 3,648 EBRT [1]	Predominantly high risk (5 of 8 studies) [1]	Metastasis-free survival (MFS) [1]	41–122.4 months (surgery) / 39.6–87 months (RT), by study [1]	6 of 8 cohorts from the United States [1]
F. H. Amo et al., 2012 [3]	No (abstract only)	Retrospective cohort [3]	435 [3]	65% RP vs. 35% RT [3]	Low and intermediate risk [3]	BCR-free survival [3]	Median 60 months (range 3–106) [3]	Not reported [3]
P. Morando et al., 2012 [4]	No (abstract only, truncated)	Retrospective, analytical, documentary [4]	139 [4]	Not specified [4]	Organ-confined, by risk group [4]	Disease-free survival until recurrence [4]	Not reported [4]	Not reported [4]
A. Martínez, 2017 [5]	No (abstract only, background only)	Not determinable from available text [5]	Not reported [5]	Not reported [5]	Not reported [5]	Not reported [5]	Not reported [5]	Not reported [5]
J. C. Velasco, 2019 [6]	No (abstract only, truncated)	Not determinable from available text [6]	Not reported [6]	RP vs. EBRT with ADT (high risk) [6]	High risk [6]	Not reported [6]	Not reported [6]	Not reported [6]
M. H. B. Dias et al., 2025 [7]	No (abstract only, metadata only)	Systematic review [7]	Not reported [7]	RP vs. RT vs. active monitoring [7]	Not reported [7]	Not reported [7]	Not reported [7]	Not reported [7]
Abraham Orozco Martínez, 2017 [2]	Yes	Randomized clinical trial [2]	1,643 (three arms) [2]	553 surgery vs. 545 RT vs. 545 active surveillance [2]	Low risk (predominantly, implied by mean PSA 4.6 ng/mL, 77% Gleason 6, 76% T1c) [2]	Mortality and progression-free survival [2]	Median 10 years [2]	Not reported [2]

The body of evidence is methodologically heterogeneous. Galdós-Bejar et al. (2024) provides the largest pooled sample through meta-analysis and is the only source that explicitly addresses robotic-assisted prostatectomy, noting it in two of its eight included cohorts [1]. The Abraham Orozco Martínez (2017) paper is the only source reporting results from a randomized trial design, based on the ProtecT trial [2]. It is notable that two 2017 publications share an identical Spanish-language title about the absence of statistically significant mortality differences between radical prostatectomy, radiotherapy, and active surveillance; "A. Martínez, 2017" is available as a background-only abstract excerpt [5], while "Abraham Orozco Martínez, 2017" provides a full text with comparative results [2] and appears to report on the same underlying work. Morando et al. (2012) [4], Velasco (2019) [6], and Dias et al. (2025) [7] contributed limited extractable data due to truncated or metadata-only abstracts, restricting their contribution to the qualitative synthesis. The patient populations across studies differ substantially: Galdós-Bejar et al. enrolled predominantly high-risk patients [3], Amo et al. restricted inclusion to low- and intermediate-risk disease [3], and the trial underlying the Orozco Martínez paper enrolled an overwhelmingly low-risk population (77% Gleason 6, 76% T1c, mean PSA 4.6 ng/mL) [2].

A critical limitation affecting the entire body of evidence with respect to the specific research question is the absence of disaggregated data on robotic-assisted radical prostatectomy (RARP). Galdós-Bejar et al. acknowledge robotic approaches in two included cohorts without isolating their outcomes [1], and no other source distinguishes robotic from open surgery [2, 3]. Consequently, the comparative evidence reviewed here predominantly concerns radical prostatectomy as a class rather than robotic-assisted surgery specifically.

Oncological Outcomes

Metastasis-Free Survival

The largest and most granular dataset on MFS is provided by Galdós-Bejar et al. (2024), whose meta-analysis of eight retrospective cohorts yielded a pooled hazard ratio estimate of 0.79 (95% CI 0.50–1.24) favoring radical prostatectomy, a difference that did not reach statistical significance [1]. Heterogeneity was extreme (I^2 = 90.34%; τ^2 = 0.31) [1], indicating that the pooled estimate should be interpreted with considerable caution. Study-level MFS rates from that review are shown below.

Cohort (within Galdós-Bejar et al., 2024)	Time point	MFS — Radical prostatectomy	MFS — EBRT (±brachytherapy)
Yamamoto [1]	10 years	73.9% [1]	88.2% [1]
Boorjian [1]	5 years	93% [1]	91% [1]
Boorjian [1]	10 years	85% [1]	81% [1]
Zelefsky [1]	8 years	97% [1]	93% [1]
Emam [1]	3 years	80% [1]	91% [1]
Emam [1]	5 years	77% [1]	91% [1]
Chul Koo [1]	5 years	66.7% [1]	58.3% [1]
Reichard [1]	5 years	89.1% [1]	88.3% [1]
Kishan — EBRT arm [1]	5 years	75% [1]	76% [1]
Kishan — EBRT + brachytherapy arm [1]	5 years	75% [1]	92% [1]
Markovina [1]	Not available	Not available [1]	Not available [1]

The direction of the treatment difference varies across individual cohorts: five cohorts show numerically higher MFS with radical prostatectomy, while two (Yamamoto and Emam) show higher MFS with radiotherapy, and two others (Reichard and the Kishan EBRT arm) are essentially equivalent [1]. The Kishan cohort is especially informative in that it compared three modalities, finding that EBRT plus brachytherapy achieved a 5-year MFS of 92%, substantially above either RP (75%) or EBRT alone (76%), in a high-risk Gleason 9–10 population [1, 1]. This modality distinction was not carried through to the pooled meta-analytic estimate, which compared RP only against EBRT without brachytherapy as the reference.

Biochemical Recurrence-Free Survival

Amo et al. (2012) is the only source providing BCR-free survival rates with explicit time-point data. Among 435 patients with low- and intermediate-risk localized prostate cancer followed for a median of 60 months, BCR-free survival rates were 84%, 75%, and 70% at 3, 5, and 7 years for radical prostatectomy versus 97%, 84%, and 64% for radiotherapy [3]. Overall BCR occurred in 22% of surgically treated and 19% of radiotherapy-treated patients, a non-significant difference (p = 0.47) [3]. Stratified analyses similarly showed no statistically significant treatment effect within the low-risk (p = 0.60) or intermediate-risk subgroup (p = 0.32) [3]. The survival curves crossed between 5 and 7 years, with radiotherapy initially yielding superior recurrence-free survival but surgery appearing to maintain a more durable response at 7 years [3, 3]. The authors explicitly cautioned that this crossover pattern meant longer follow-up could alter the conclusions [3].

The BCR definition used for each arm was not stated in the available abstract [3], which is a material limitation, since the PSA threshold for recurrence after surgery (typically ≥0.2 ng/mL) differs substantially from the Phoenix criterion used after radiotherapy (PSA nadir +2 ng/mL). The two definitions are not directly comparable, and this methodological ambiguity limits interpretation of the BCR equivalence finding.

Overall Survival and Cancer-Specific Mortality

The Abraham Orozco Martínez (2017) RCT data, which described 17 prostate cancer-related deaths across all three arms at a median follow-up of 10 years, found no statistically significant difference in cancer-specific mortality by assigned treatment group (p = 0.48) [2]. Long-term survival and progression-free survival were described as equivalent between surgery and radiotherapy, particularly in low-risk patients [2, 2]. Galdós-Bejar et al. (2024) did not report OS or CSS as outcomes of its included comparisons [1], and no other source provided arm-specific OS or CSS estimates [3–7].

The table below summarizes the available quantitative oncological outcomes across all contributing sources.

Source	Endpoint	RP result	RT result	Comparative estimate	p-value / significance
Galdós-Bejar et al., 2024 [1]	MFS (pooled)	Lower metastasis risk	Reference	HR 0.79 (95% CI 0.50–1.24) [1]	Not significant; I^2=90.34% [1]
F. H. Amo et al., 2012 [3]	BCR-free survival at 3 years	84% [3]	97% [3]	Not reported [3]	p=0.47 overall [3]
F. H. Amo et al., 2012 [3]	BCR-free survival at 5 years	75% [3]	84% [3]	Not reported [3]	p=0.47 overall [3]
F. H. Amo et al., 2012 [3]	BCR-free survival at 7 years	70% [3]	64% [3]	Not reported [3]	p=0.47 overall [3]
Abraham Orozco Martínez, 2017 [2]	Cancer-specific mortality	No significant difference [2]	No significant difference [2]	Not reported numerically [2]	p=0.48 [2]
Abraham Orozco Martínez, 2017 [2]	Progression-free survival	Similar (low risk) [2]	Similar (low risk) [2]	Not reported numerically [2]	Not significant [2]

Functional Outcomes and Toxicity

Comparative functional outcome data are sparse across the included sources. The only source providing time-stratified comparisons of functional morbidity is the Abraham Orozco Martínez (2017) RCT, which reported that

patients treated with radical prostatectomy experienced more urinary incontinence, erectile dysfunction, and urinary urgency than those treated with radiotherapy at both 2 and 5 years of follow-up, while no significant difference in functional complications was observed at 15 years [2, 2]. No rates, severity grades, or validated instrument scores were available from the full text of that source [2].

Galdós-Bejar et al. (2024) described the toxicity profiles of the two modalities in qualitative terms only: radical prostatectomy was associated with predominantly genitourinary toxicity, particularly urinary incontinence, while radiotherapy was associated with gastrointestinal toxicity such as rectal bleeding, with overall surgical toxicity described as tending to be greater [1, 1]. No incidence rates, severity grading by CTCAE or RTOG criteria, or time-point data were reported [1]. All eight cohorts included in that meta-analysis administered radiotherapy in combination with ADT [1], yet no data on ADT-related adverse effects (metabolic, cardiovascular, or other) were reported [1].

None of the remaining sources provided arm-specific toxicity rates, complication grading, or validated quality-of-life instrument scores [3–7]. The absence of instrument-based QoL measurement across this evidence base means that the differential functional burden of surgery versus radiotherapy, particularly with respect to robotic prostatectomy, cannot be quantitatively characterized from the available literature.

Treatment Modalities

For radical prostatectomy, retropubic, open, and robotic-assisted approaches were noted across the cohorts included in Galdós-Bejar et al. (2024), with pelvic lymph-node dissection performed in three of the eight studies and robotic-assisted prostatectomy reported in two; nerve-sparing technique and standard versus extended lymph-node dissection were not characterized [1]. The remaining sources described radical prostatectomy without specifying the approach or lymph-node management [2, 3].

For radiotherapy, EBRT was the modality used in all eight cohorts of the Galdós-Bejar et al. meta-analysis, with a reported standard dose above 70 Gy and no fractionation scheme specified; one study (Kishan) additionally compared EBRT combined with brachytherapy as a third arm [1]. ADT was administered alongside radiotherapy in all included studies, with rates per EBRT arm ranging from 56% to 100% across individual cohorts [1]. The remaining primary studies did not specify the type or technique of radiotherapy used [2, 3].

Synthesis

The available evidence converges on a finding of no statistically significant difference in oncological outcomes between radical prostatectomy and radiotherapy for localized prostate cancer, whether measured as BCR-free survival, MFS, or cancer-specific mortality. However, three factors critically limit the validity of treating this null result as a population-level equivalence statement.

Population heterogeneity and risk-group mismatch. The two most data-rich sources in this review drew from dramatically different patient populations: the meta-analysis by Galdós-Bejar et al. enrolled predominantly high-risk patients (five of eight cohorts) [1], while the RCT data reported by Orozco Martínez enrolled a predominantly low-risk population (77% Gleason 6, 76% T1c, mean PSA 4.6 ng/mL) [2], and Amo et al. explicitly restricted inclusion to low- and intermediate-risk disease [3]. The absence of a statistically significant MFS advantage for surgery in a high-risk cohort [1] and the absence of a cancer-specific mortality advantage in a low-risk trial [2] are both credible findings within their respective populations, but they cannot be pooled into a single inference. The Kishan cohort, embedded within the Galdós-Bejar meta-analysis, further complicates the comparison by demonstrating that EBRT combined with brachytherapy achieved meaningfully higher 5-year MFS (92%) than either EBRT alone (76%) or RP (75%) in a Gleason 9–10 high-risk population [1], underscoring that ”radiotherapy” as a category encompasses substantially different treatment intensities.

Confounding by ADT use. In every radiotherapy arm within the Galdós-Bejar et al. meta-analysis, patients received concurrent ADT [1], yet RP patients did not receive routine ADT as primary treatment. This represents a systematic imbalance between arms that cannot be resolved through retrospective adjustment. The authors acknowledged covariate assessment was not uniform across cohorts and listed ADT use, comorbidities, age, and risk category as

uncontrolled potential confounders [1, 1]. The apparent equivalence in MFS—and in some cohorts even numerical superiority of the radiotherapy arm (Yamamoto, Emam) [1] —may therefore reflect the additive effect of ADT rather than the radiation itself.

Methodological hierarchy and the unique role of the RCT. The majority of quantitative data derive from retrospective cohorts with known susceptibility to selection bias: patients selected for surgery or radiotherapy are not equivalent at baseline in clinical practice, and the included studies did not uniformly adjust for the relevant confounders [1]. The Orozco Martínez (2017) RCT offers the only randomized comparison and, by virtue of its design, provides the strongest causal evidence, yet its population (predominantly low-risk, mean PSA 4.6 ng/mL, 10-year median follow-up) [2, 2] is not representative of high-risk disease where the surgical versus radiotherapy question is most contentious and where the retrospective literature is most concentrated. The BCR crossover pattern observed by Amo et al.—radiotherapy superior at 3 and 5 years, surgery appearing more durable at 7 years [3, 3] —suggests that follow-up duration substantially shapes the apparent comparative benefit, and that the 10-year RCT follow-up, though the longest in this body of evidence, may still be insufficient to detect late divergence in oncological outcomes in higher-risk disease.

Taken together, these considerations support the following risk-stratified interpretive framework: in low-risk and favorable intermediate-risk disease, the available randomized evidence supports equivalence of radical prostatectomy and radiotherapy with respect to cancer-specific mortality, with the differential functional toxicity profile favoring radiotherapy in the first decade (less incontinence and erectile dysfunction) but not necessarily at 15 years [2, 2]. In high-risk and very high-risk disease, the retrospective meta-analytic evidence neither confirms nor excludes a clinically meaningful benefit of either modality at the 5-year MFS horizon [1, 1], and the addition of brachytherapy to EBRT may confer a distinct advantage over RP in the highest-grade tumors [1]. The specific contribution of robotic versus open prostatectomy to these outcomes remains entirely uncharacterized from the included literature [1–3].

References

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2. Martínez AO (2017) Prostatectomía radical, radioterapia y vigilancia activa sin diferencias estadísticamente significativas en los resultados de mortalidad de pacientes con cáncer de próstata localizado. Revista Mexicana de Urología. https://doi.org/10.48193/rmu.v77i2.12

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